

A Geometry Problem

Problem statement

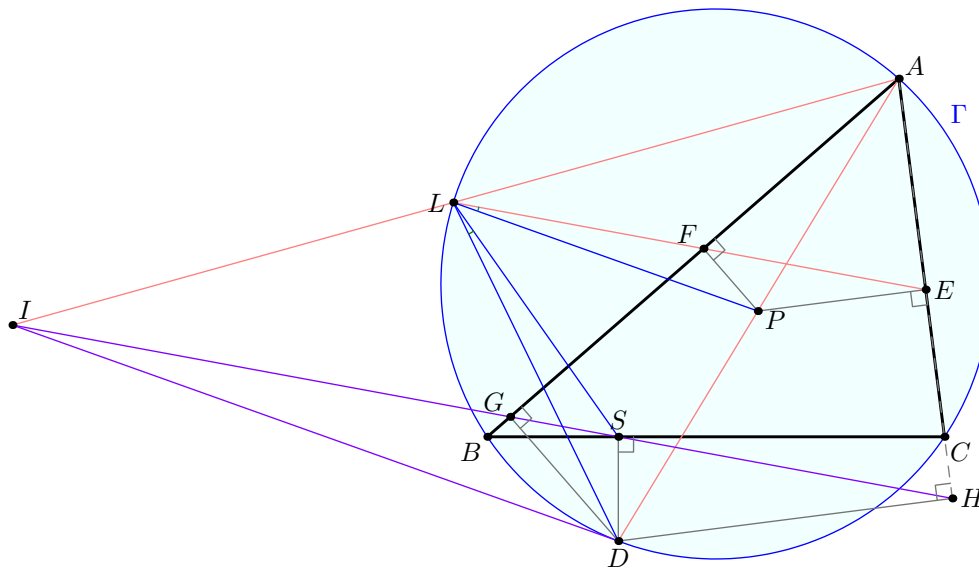
Source: We Love Geometry WeChat public account, Teacher Yao Jiabin.

Let $\triangle ABC$ be inscribed in a circle Γ . Let P be an interior point of $\triangle ABC$.

Let E and F be the feet of the perpendiculars from P to AC and AB , respectively. Let L be the intersection of line EF with Γ shown in the figure. Let line AP meet Γ again at D . Finally, let S be the foot of the perpendicular from D to BC .

Prove that

$$\angle PLE = \angle SLD.$$



Proof. Let G and H be the feet of the perpendiculars from D to AB and AC , respectively. Thus

$$DG \perp AB, \quad DH \perp AC.$$

Since D lies on the circumcircle of $\triangle ABC$, by the Simson line theorem, the three feet G, S, H are collinear. All ratios of collinear segments below are understood as directed ratios.

Let

$$I := AL \cap GH.$$

We first claim that $LP \parallel ID$. Indeed, since

$$PF \perp AB, \quad DG \perp AB,$$

we have $PF \parallel DG$. Hence

$$\triangle AFP \sim \triangle AGD,$$

and therefore

$$\frac{AF}{AG} = \frac{AP}{AD}.$$

Similarly, since

$$PE \perp AC, \quad DH \perp AC,$$

we have $PE \parallel DH$. Hence

$$\triangle AEP \sim \triangle AHD,$$

and thus

$$\frac{AE}{AH} = \frac{AP}{AD}.$$

Consequently,

$$\frac{AF}{AG} = \frac{AE}{AH},$$

which implies

$$EF \parallel GH.$$

Since L, E, F are collinear and I, G, H are collinear, this gives

$$LE \parallel IH.$$

Moreover, applying the intercept theorem in triangle AIG , with $L \in AI$, $F \in AG$, and $LF \parallel IG$, gives

$$\frac{AL}{AI} = \frac{AF}{AG}.$$

Therefore

$$\frac{AL}{AI} = \frac{AP}{AD}.$$

Since A, L, I are collinear and A, P, D are collinear, it follows that

$$LP \parallel ID.$$

Now

$$LE \parallel IH, \quad LP \parallel ID, \quad EP \parallel HD.$$

Hence

$$\triangle ELP \sim \triangle HID.$$

In particular,

$$\angle PLE = \angle DIH.$$

It remains to prove that

$$\angle DIH = \angle DLS.$$

Since A, L, D, C are concyclic and A, L, I are collinear, we have

$$\angle ILD = \angle ALD = \angle ACD.$$

Also,

$$DS \perp BC, \quad DH \perp AC,$$

so

$$\angle DSC = \angle DHC = 90^\circ.$$

Thus S, C, H, D are concyclic. Hence

$$\angle HSD = \angle HCD.$$

Since A, C, H are collinear, we have

$$\angle ACD = \angle HCD.$$

Since I, S, H are collinear, we have

$$\angle ISD = \angle HSD.$$

Therefore

$$\angle ACD = \angle ISD.$$

Together with $\angle ILD = \angle ACD$, this gives

$$\angle ILD = \angle ISD.$$

Thus I, L, D, S are concyclic.

It follows that

$$\angle DLS = \angle DIS.$$

Since I, S, H are collinear, we have

$$\angle DIS = \angle DIH.$$

Combining this with $\angle PLE = \angle DIH$, we obtain

$$\angle PLE = \angle DLS.$$

Therefore, in ordinary angles,

$$\angle PLE = \angle SLD.$$

This proves the desired result. □